Course: [BCS201] Programming in C (Programming for Problem Solving)

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: FORMULA SHEET**

Module / Unit: Full Syllabus **Faculty Author: Department of Computer Scie**

C Language Quick Syntax & Standard Library Cheatsheet

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1. Pointers & Memory Management Cheatsheet

- Pointer Operators: & (address-of), * (dereference / value-at-address).
- Dynamic Allocation: malloc(size), calloc(n, size), realloc(ptr, size), free(ptr).
- Function Pointers: Syntax, callbacks, and dispatch tables.

2. Standard I/O and String Functions

- String Operations: strlen(), strcpy(), strncpy(), strcat(), strcmp(), strstr().
- File Streams: fopen("mode"), fclose(fp), fprintf(), fscanf(), fgets(), fputs(), fread(), fwrite().
- File Position: fseek(), ftell(), rewind() with SEEK_SET, SEEK_CUR, SEEK_END.